



KCED
Learning Augmented

24ADI003-MACHINE LEARNING

LINEAR REGRESSION OBJECTIVE

- Learn a function that best explains the relationship between inputs and output.

Graphically:

- **Points = data**
- **Curve / line = model**
- Linear Regression is:
 - Simplest
 - Interpretable
 - Has closed-form math
 - Foundation of all regression models

The **linear regression model** is written as:

$$y = w_0 + w_1x$$

Prediction = base value + (effect × input)

LINEAR REGRESSION OBJECTIVE

- **Example:** Auto fare rule:

$$\text{Fare} = \text{Base charge} + (\text{cost per km} \times \text{distance})$$

Auto Fare	Linear Regression
Base charge	(w_0) (bias)
Cost per km	(w_1) (weight)
Distance	(x)
Fare	(y)

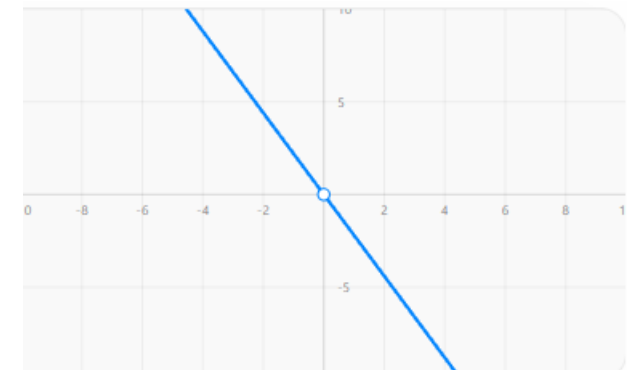
LINEAR REGRESSION OBJECTIVE

Symbol	Meaning	Description
(x)	Input / Feature	What we know (study hours, experience, area)
(y)	Output / Target	What we want to predict (marks, salary, price)
(w_1)	Weight / Slope	How strongly x affects y
(w_0)	Bias / Intercept	Starting value of y

When you draw the equation on a graph:

- x-axis $\rightarrow$ input (study hours)
- y-axis $\rightarrow$ output (marks)
- The equation draws a **straight line**

That's why it's called **linear** regression.



$$\text{Prediction} = \text{base value} + (\text{effect} \times \text{input})$$

LINEAR REGRESSION

WEIGHT w_1 (Slope)

Definition

w_1 tells us:

How much y changes when x increases by 1 unit.

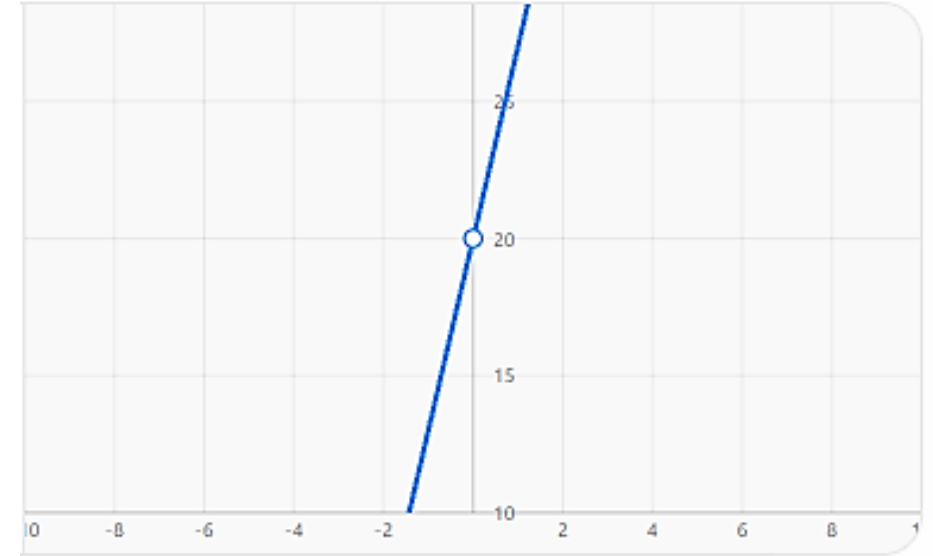
$$y = 20 + 7x$$

Then:

- Increase study hours by 1 → Marks increase by 7

Study Hours	Predicted Marks
1	27
2	34
3	41

The line goes upward → positive slope.
Sign of the Weight



(w_1) value	Meaning
$(w_1 > 0)$	As $x \uparrow$, $y \uparrow$
$(w_1 < 0)$	As $x \uparrow$, $y \downarrow$
$(w_1 = 0)$	x has no effect

LINEAR REGRESSION

BIAS w_0 (Intercept)

Definition:

w_0 is the value of y when $x = 0$.

$$y = 20 + 7x$$

If:

- $x = 0$ hours studied

Then:

$$y = 20$$

So even without studying, the model predicts **20 marks**.

Why Bias Is Needed

Imagine no bias:

$$y = 7x$$

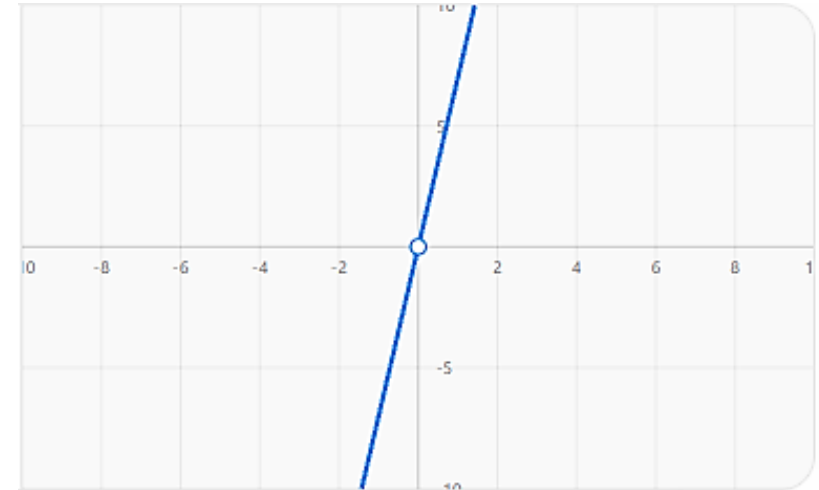
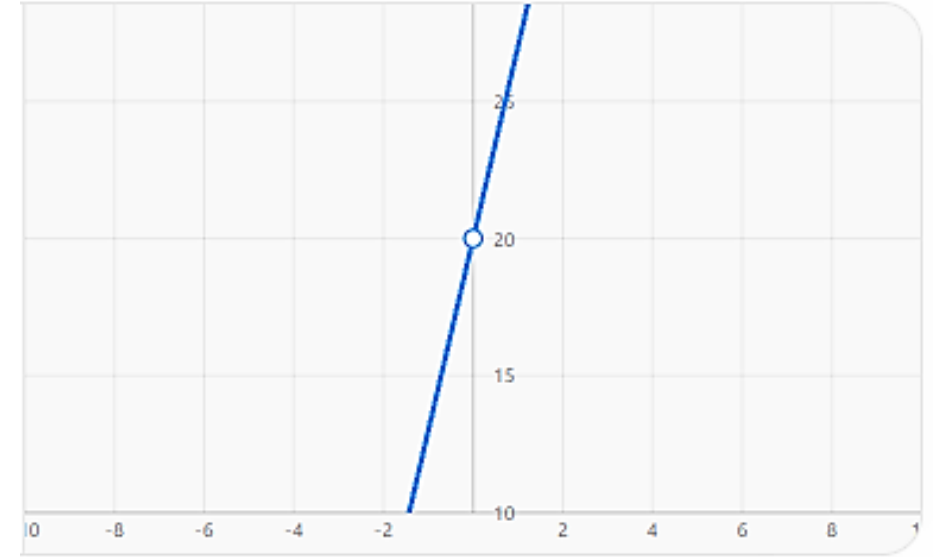
Then at $x = 0$:

- $y = 0$ (always)

But real life rarely starts from zero.

- Students may score marks without studying
- Salary exists even with 0 experience
- House has value even with 0 renovation

Bias shifts the line up or down to fit reality.



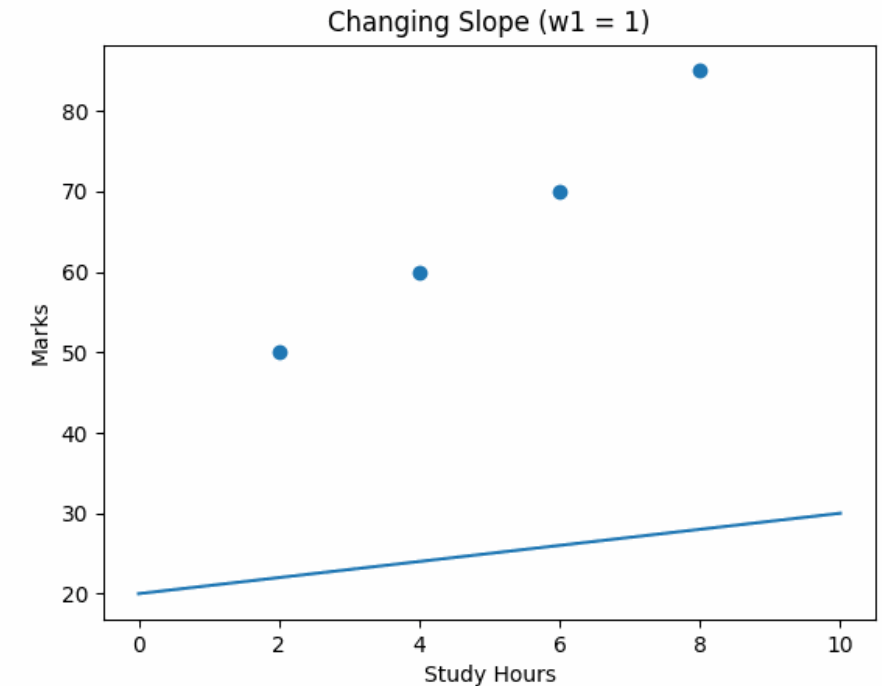
LINEAR REGRESSION

How the Line Changes When Weights Change: Case 1: Change ONLY w_1 (Slope)

Effect	What happens
Increase (w_1)	Line becomes steeper
Decrease (w_1)	Line becomes flatter

The importance of x increases or decreases.

- Increasing $w_1 \rightarrow$ line tilts more steeply
- Decreasing $w_1 \rightarrow$ line flattens
- Interpretation: *importance of input feature x changes*



LINEAR REGRESSION

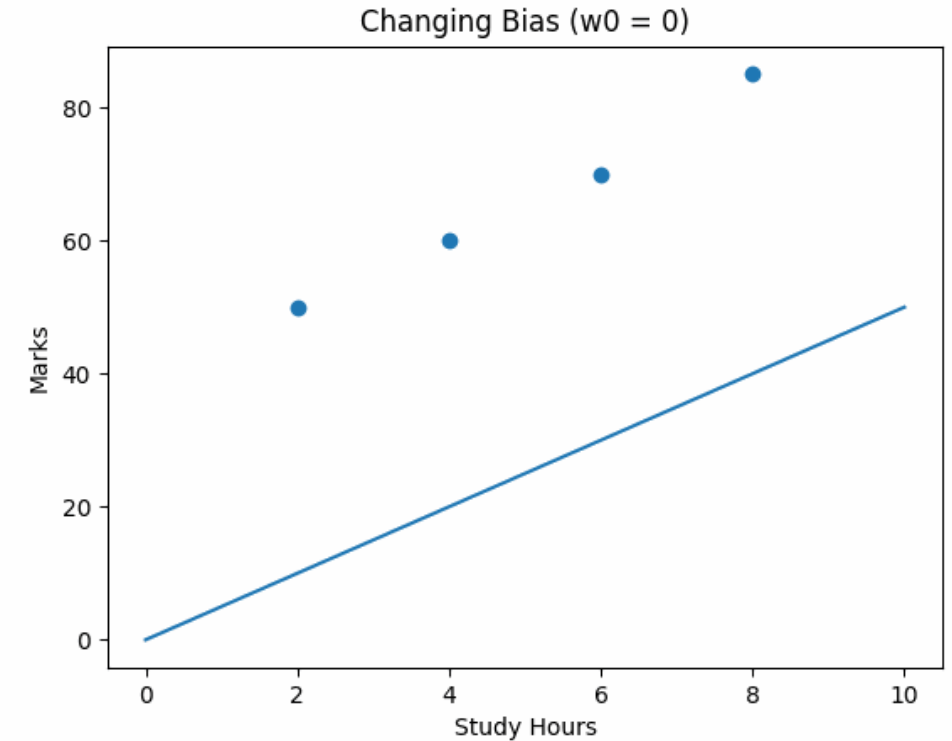
How the Line Changes When Weights Change:

Case 2: Change ONLY w_0 (Bias)

Effect	What happens
Increase (w_0)	Line moves up
Decrease (w_0)	Line moves down

base prediction changes.

- Increasing $w_0 \rightarrow$ line shifts upward
- Decreasing $w_0 \rightarrow$ line shifts downward
- Interpretation: *baseline prediction changes*



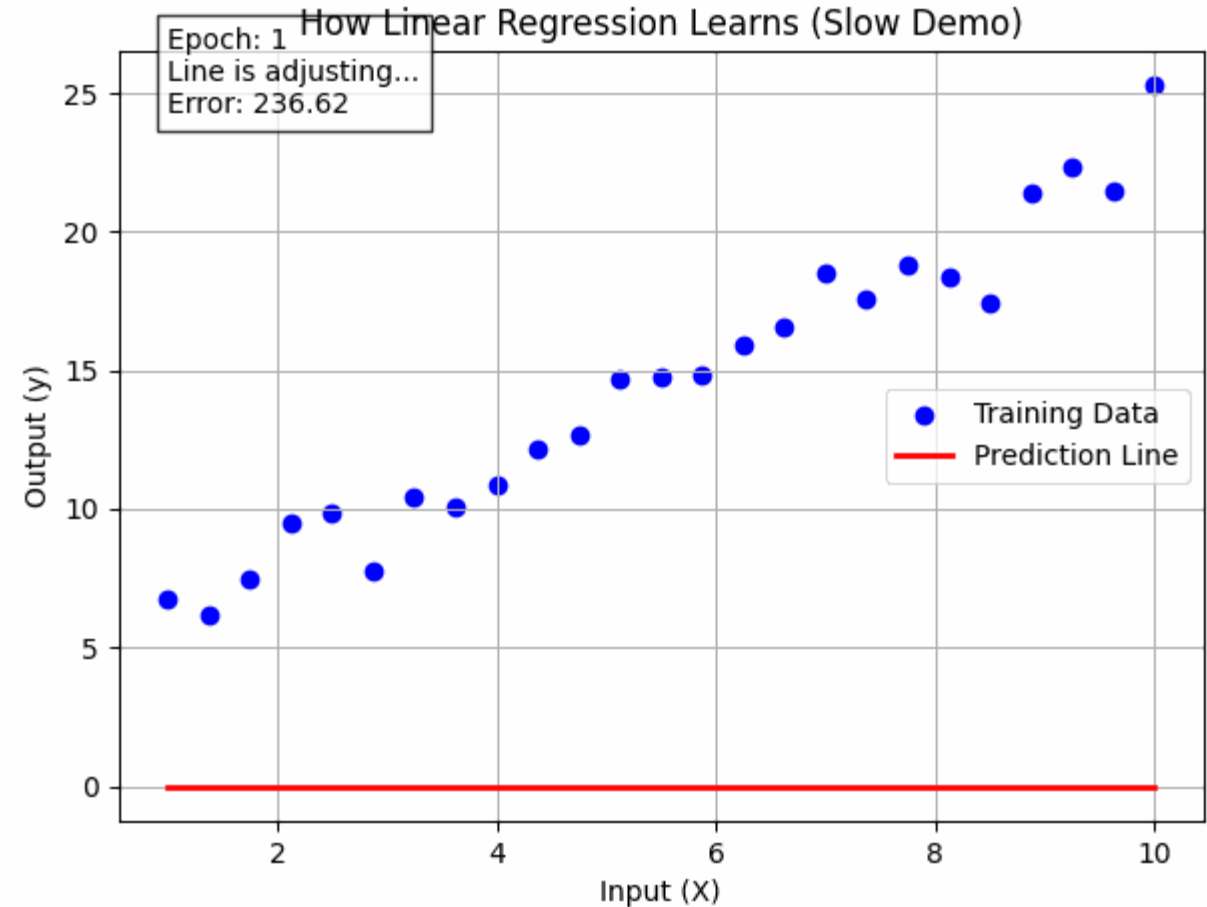
LINEAR REGRESSION

In ML training:

- We **do not fix** w_0, w_1
 - The algorithm **learns** them from data
 - Each update changes the line
- So:

Changing weights = changing the model itself

This is why your infographic shows the line moving.



LINEAR REGRESSION

Why “Best Line”? → Least Squares Idea

For each data point:

$$\text{Error} = y_i - \hat{y}_i$$

If we simply add errors, positives and negatives cancel

So we **square them**:

$$\text{Squared Error} = (y_i - \hat{y}_i)^2$$

Objective Function (Cost Function)

Least Squares Estimator finds w_0, w_1 that minimize this sum.

LINEAR REGRESSION

Example

Given Dataset (Study Hours vs Marks)

Student	x (Hours)	y (Marks)
1	1	50
2	2	55
3	3	65
4	4	70
5	5	75

Step 1: Compute Required Sums

Quantity	Calculation	Value
(n)	—	5
($\sum x$)	1+2+3+4+5	15
($\sum y$)	50+55+65+70+75	315
($\sum x^2$)	1+4+9+16+25	55
($\sum xy$)	50+110+195+280+375	1010

LINEAR REGRESSION

Example

Step 2: Calculate Slope w_1

$$w_1 = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

$$w_1 = \frac{5(1010) - 15(315)}{5(55) - 15^2} = \frac{5050 - 4725}{275 - 225} = \frac{325}{50} = 6.5$$

Step 3: Calculate Bias w_0

$$w_0 = \frac{\sum y - w_1 \sum x}{n}$$

$$w_0 = \frac{315 - 6.5(15)}{5} = \frac{315 - 97.5}{5} = 43.5$$

Final Regression Equation

$$\hat{y} = 43.5 + 6.5x$$

LINEAR REGRESSION

Example

Step 4: Prediction

If student studies **6 hours**:

$$\hat{y} = 43.5 + 6.5(6) = 82.5$$

Slope $w_1 = 6.5$

Each additional hour of study increases marks by **~6.5**

Bias $w_0 = 43.5$

Baseline marks even with **zero study**

Maximum Likelihood Estimation (MLE) in Linear Regression

“Likelihood”:

Likelihood answers this question:

If I choose a particular line, how likely is it that this line generated my data?

Bad line → data very unlikely

Good line → data very likely

MLE chooses:

Parameters that maximize likelihood

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Maximum Likelihood Estimation (MLE) in Linear Regression

Why do we even need MLE?

In real life:

Data is **never perfect**

Measurements have **noise**

The same input x can produce slightly different outputs y

So instead of saying

“This line is correct”

We say

“This line is **most likely** to have produced the data”

That idea is **Maximum Likelihood Estimation**.

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Maximum Likelihood Estimation (MLE) in Linear Regression

The Linear Regression Model (Starting Point)

We assume:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

Where:

β_0 → intercept

β_1 → slope

ε_i → error (noise)

We do NOT assume error is zero

We assume error is **random**

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Maximum Likelihood Estimation (MLE) in Linear Regression

Critical Assumption of MLE (Very Important)

Errors are:

Normally distributed

Mean = 0

Symmetric

Same variance σ^2

Interpretation:

Positive and negative errors are equally likely

No systematic bias

Most errors are small

Big errors are rare

This leads to the bell curve

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Maximum Likelihood Estimation (MLE) in Linear Regression

Mathematical Form of Likelihood

For one data point:

$$P(y_i | x_i, \beta_0, \beta_1) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \hat{y}_i)^2}{2\sigma^2}\right)$$

Where:

$$\hat{y}_i = \beta_0 + \beta_1 x_i$$

Probability depends on **squared error**

Log-Likelihood (Why we take log)

Multiplying many probabilities is hard.

So we take **log**.

Log-likelihood becomes:

Maximizing likelihood $\Leftrightarrow$ Minimizing squared error

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Maximum Likelihood Estimation (MLE) in Linear Regression

EXAMPLE 1

x	y
1	27
2	34
3	41

Step 1: Assume a regression model

$$y = \beta_0 + \beta_1 x$$

Step 2: Try a candidate line

Let us test:

$$y = 20 + 7x$$

Maximum Likelihood Estimation (MLE) in Linear Regression

EXAMPLE 1

Step 3: Compute predictions and errors

x	Actual y	Predicted y	Error $e = y - \hat{y}$	e^2
1	27	27	0	0
2	34	34	0	0
3	41	41	0	0

Step 4: Likelihood interpretation

Errors are zero

Squared error sum = 0

Likelihood is **maximum**

This line is **extremely likely**

Maximum Likelihood Estimation (MLE) in Linear Regression

EXAMPLE 2

$$y = 10 + 5x$$

Step 1: Predictions and errors

x	Actual y	Predicted y	Error	(e ²)
1	27	15	12	144
2	34	20	14	196
3	41	25	16	256

Step 2: Total squared error

$$144 + 196 + 256 = 596$$

Step 3: Likelihood interpretation

Errors are large

Squared errors are huge

Likelihood becomes very small

This line is **very unlikely**

Step 4: MLE decision

$$\text{MLE chooses } y = 20 + 7x$$

Because it maximizes likelihood

Maximum Likelihood Estimation (MLE) in Linear Regression

Why squared errors specifically?

Because of **symmetric Gaussian errors**:

+5 error is as bad as -5

Squaring removes sign

Larger mistakes punished more

Mathematically, **Gaussian distribution leads exactly to squared loss.**

Theorem:

If errors are normally distributed, the Maximum Likelihood Estimator of linear regression parameters is identical to the Ordinary Least Squares estimator.

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

Maximum Likelihood Estimation (MLE) in Linear Regression

Estimating Noise Variance

Once we get the best line:

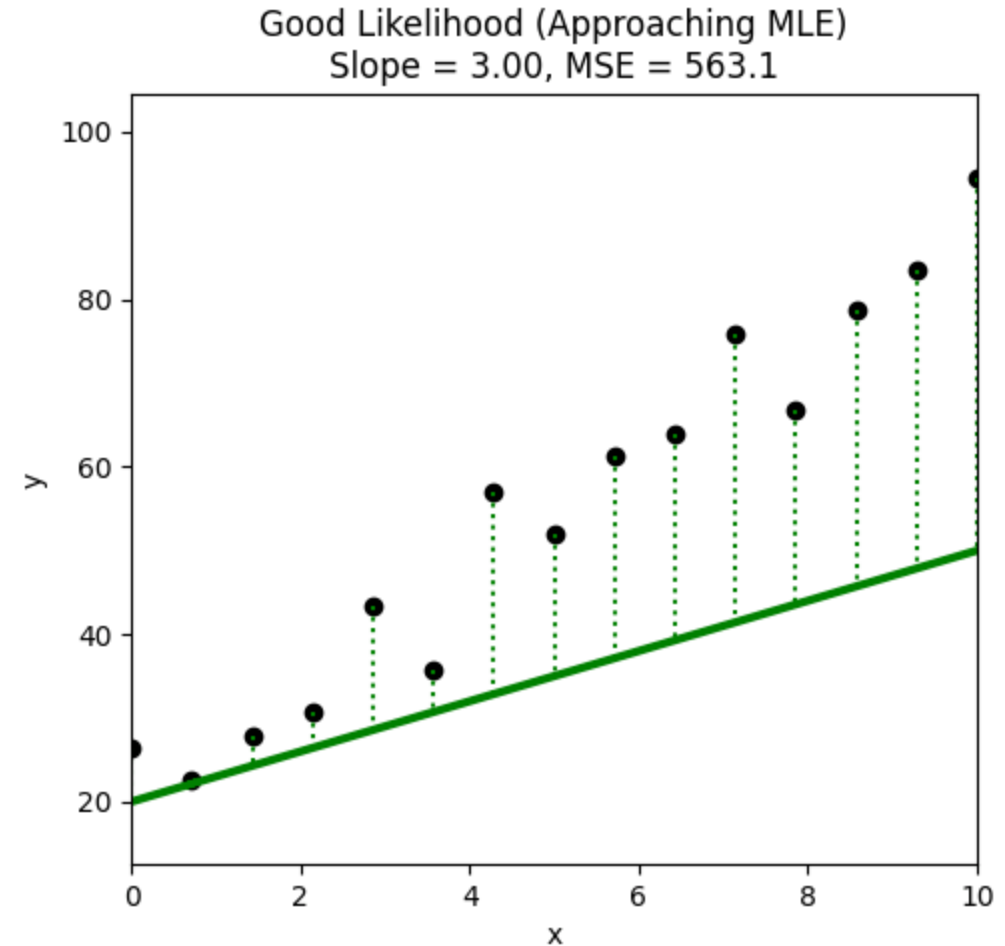
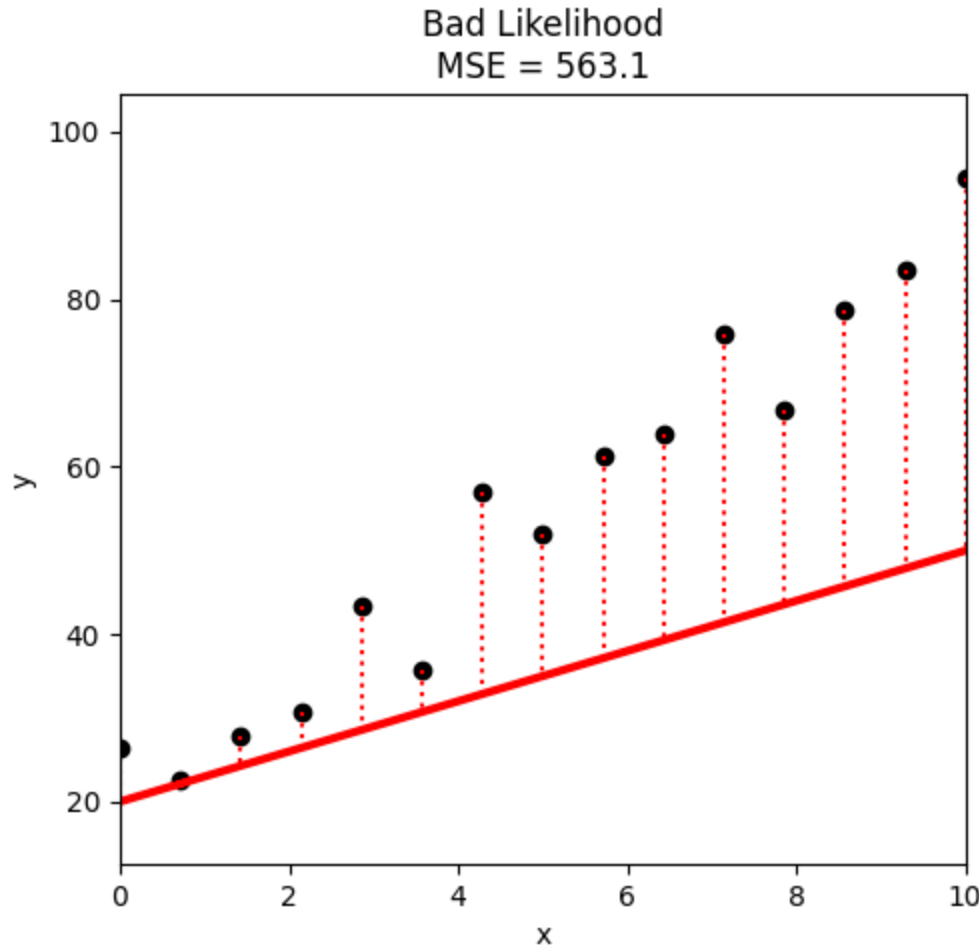
Interpretation:

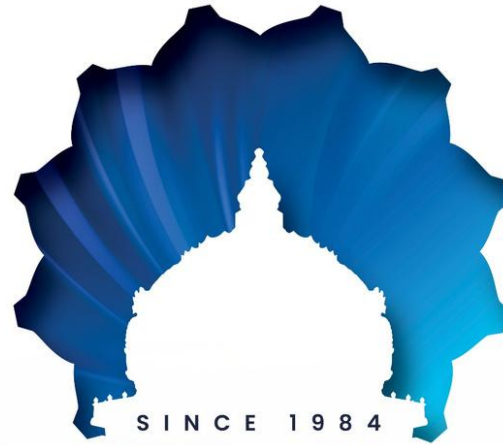
Small $\sigma^2 \rightarrow$ tight data

Large $\sigma^2 \rightarrow$ noisy data

- Data has noise
- Noise is assumed normal and symmetric
- Likelihood measures how probable data is
- MLE maximizes likelihood
- Maximizing likelihood = minimizing squared errors
- Therefore, MLE = Least Squares

Maximum Likelihood Estimation (MLE) in Linear Regression





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